

NeuroGecko

A biologically-grounded leopard gecko (*Eublepharis macularius*) in MuJoCo. Every number is tagged **PUBLISHED**, **DERIVED** or **INVENTED**, and every hypothesis that was ever tested stays in the record — especially the wrong ones. This document is that record, drawn.

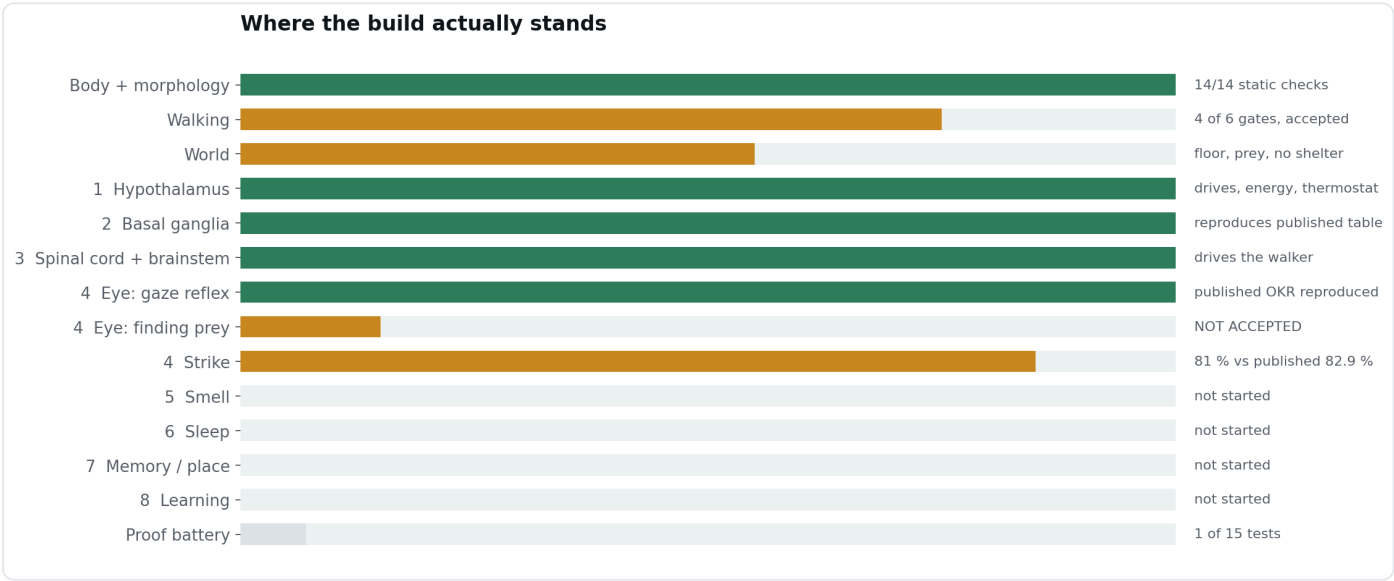
154
REFUTED

21
CONFIRMED

183
HYPOTHESES TESTED

92
COMMITTS

84% of everything tried was wrong. That is not the failure of the project — it is the product of it. A refuted entry is a map of where not to look, and nothing is ever deleted from it.



Charts and tables in this document are generated by `tools/build_report.py` from `docs/FAILURE_MAP.md`, `git log` and the evidence JSON. No figure is transcribed by hand.

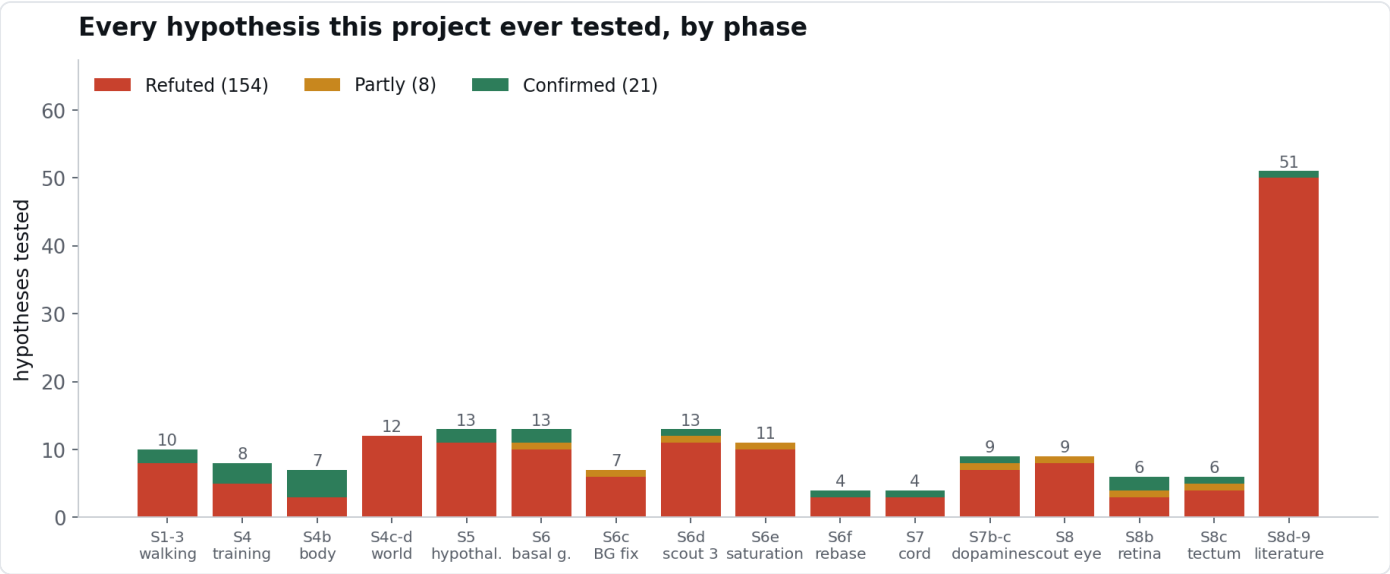
The road map

● working, checked ● partial ● built and not accepted ● not started

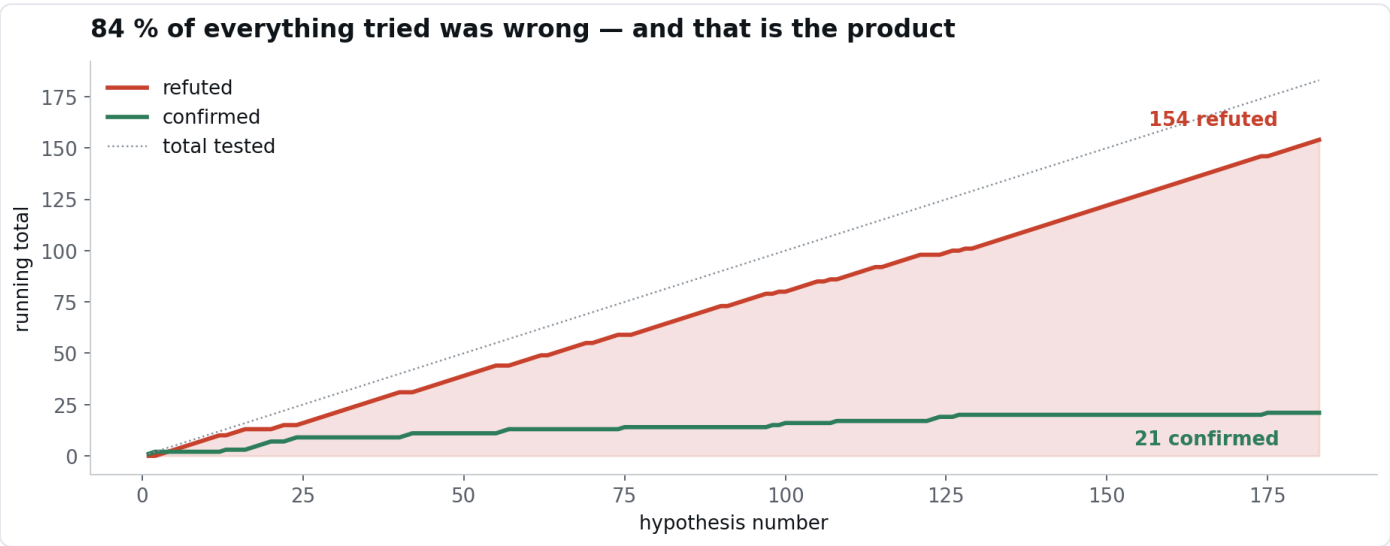


Eight brain modules. Five stand. One is built and openly recorded as not working. Three have not been started, and two of those three have such thin published evidence that building them would mean inventing.

Every hypothesis, every phase



The tall bar on the right is the current phase: a literature sweep that refuted more assumptions than any build session.



Refutations accumulate; confirmations barely move. A project that confirmed most of its guesses would be one that was not testing them.

Patterns — the same mistake, again

Six failure shapes account for most of what went wrong. Each one recurred after being “fixed”, which is why they are worth naming rather than merely counting.

×4 A default won because nobody named the parameter

The goal oracle (#165), the food oracle, the gait profile (#177), and twice before that (#26, #33). Each time a constructor omitted an argument and the callee's default silently decided the science.

Now: Every one is now passed explicitly and reported in `info` on every step.

×3 The test compared the code against itself

A class literally named *the detector cannot disagree with the world* read the prey colour from the model, built a patch of that colour, and checked the detector saw it. It never touched the renderer. It passed while the detector scored 0.000 on the real prey — twice, three sessions apart.

Now: Assertions now reach the thing being guarded, not a copy of the input.

×2 A fix that provably could not do anything

The tectum subtracted a scalar from the motion map to reject self-motion. `argmax(clip(x-c,0,None)) == argmax(x)` for any scalar `c` — identical on 20 000 of 20 000 random maps. It had never changed one reported bearing.

Now: Replaced with a spatially varying surround; it moves the answer on 1 735/2 000.

×3 A number wearing another species' clothes

`cost_of_transport = 0.73` was tagged *E. macularius*, *verified*. It is *Teratoscincus przewalskii* — a different family. The scorecard had it right; the machine-readable registry the code reads did not.

Now: Corrected, and the gap recorded: no cost of transport has ever been measured in this species.

×3 I explained away a real observation three times

The user said the walk dragged. I answered: camera shake, then half-speed playback, then an under-actuated forelimb they would have to accept. All three were wrong about the cause.

Now: The limp was the trained policy — a policy this project had already measured at 3/6 against the base's 4/6 and rejected (#183).

×2 The map already contained the answer

#71 states outright that the accepted walker IS the open-loop base. #11 measured the trained residual as worse. The evidence file records its controller as *zero residual*. I read all three and still loaded the checkpoint.

Now: Reading the ledger is not the same as believing it.

Failure → what it became

The dopamine curve sat 0.10 too high

WAS

Three sessions of a basal-ganglia model that would not match the published table.

WHY

The mechanism was wrong in the opposite direction to the previous correction. Afferent gain ($1 \pm \lambda$) with threshold 0.2.

NOW

Mean error 10.3 pp → **0.103 pp**, axis slope 1.000.

The gecko could not see its own food

WAS

The detector tested for green; the prey had been changed to brown. Exactly 0.000 at every illumination.

WHY

Read the colour the renderer actually draws — material first, not just geom rgba.

NOW

Two separate occurrences fixed, and the self-referential test replaced.

The eye could not find the cricket

WAS

Four eye configurations, 64→512 px. Correlation with true prey bearing ≈ 0 in all four.

WHY

It was never the eye. The prey travelled **0.0 mm** and was off-image 81 % of frames.

NOW

Prey now walks 398 mm per episode. The eye has something to look at for the first time.

Eight gate contracts failing for four sessions

WAS

Reported as eight separate failures; the walking evidence looked unverifiable.

WHY

One body check compared raw bytes and raised before the others could run. What differed was a *comment*.

NOW

All 16 pass. The physics hash matched all along: `b178bf26...`

Pursuit could never work

WAS

The hunting channel had no route to success at any parameter value.

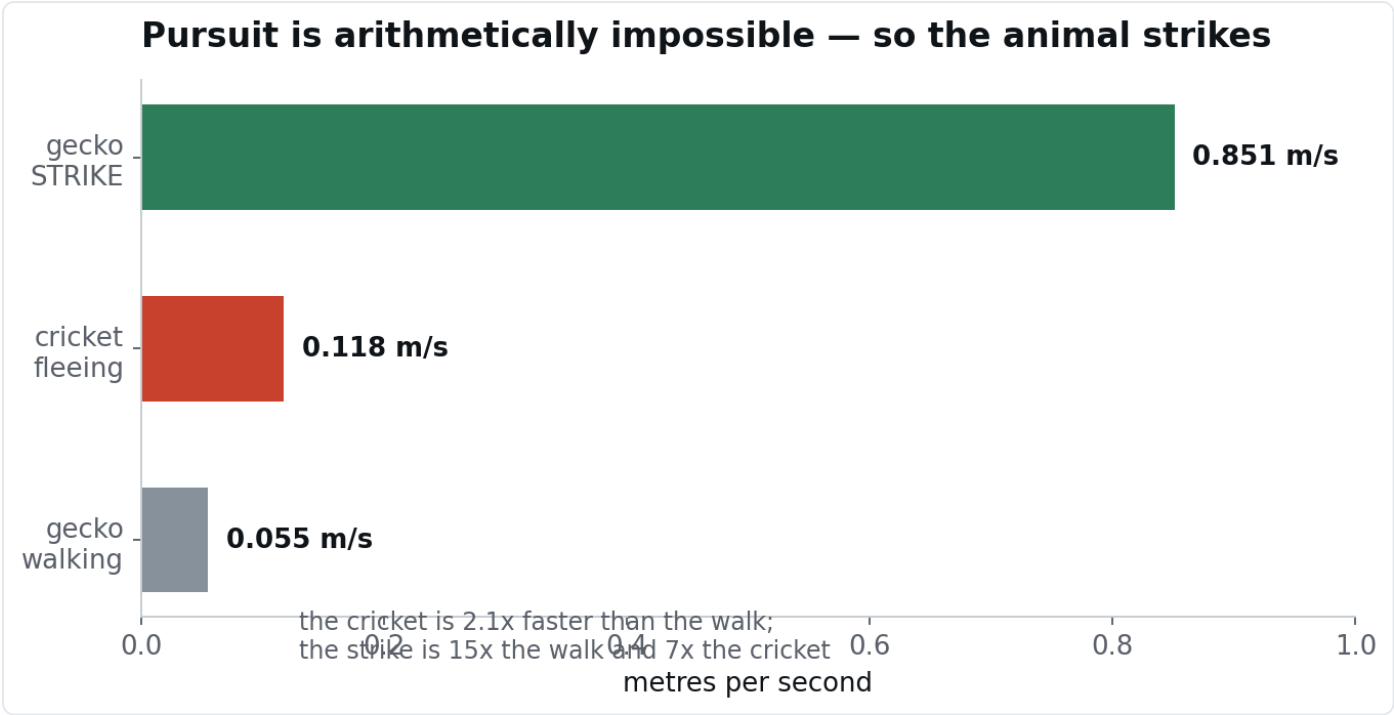
WHY

Walk 0.055 m/s, cricket 0.118 m/s. The real animal does not chase — it strikes from 2 cm at 0.851 m/s.

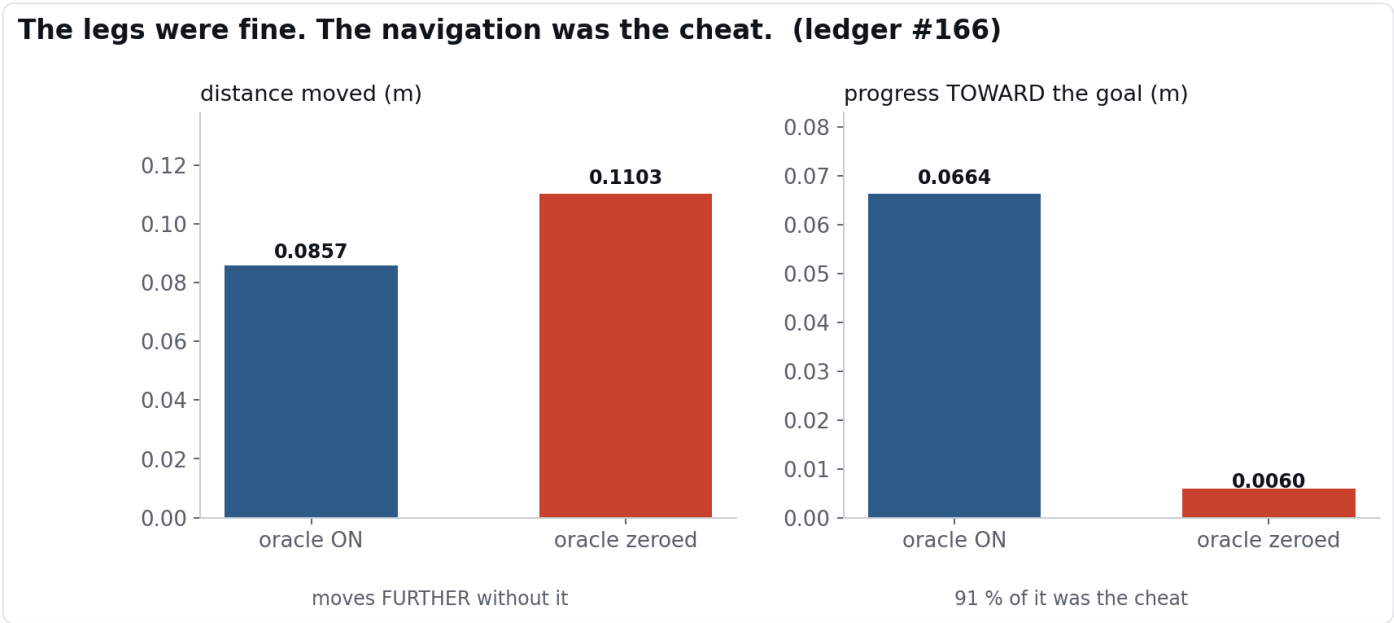
NOW

Strike built: **81 % capture** against a published 82.9 %, unfitted.

Three measurements worth keeping

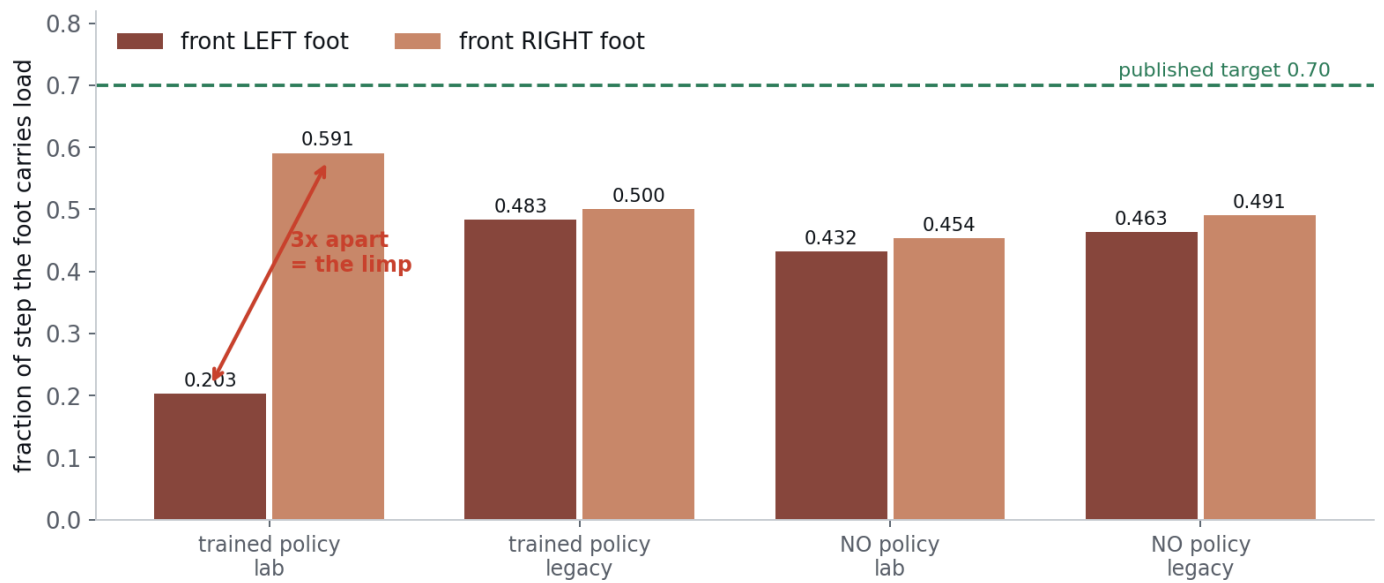


The arithmetic that forced a strike to exist.



Zeroing the privileged goal channel: the animal moves further and arrives nowhere.

The limp was the trained policy, not the body (ledger #183)



The defect the user found by watching the animal move, then measured. The trained policy limps; the hand-written base does not.

What is next

- 1 Retrain the walker for precision approach** **BLOCKING**
It arrives within 4 cm of a goal and its nose trails its trunk by 5 cm. The strike needs 2 cm. Nothing in the hunt can complete until this closes.
- 2 Settle whether lab or legacy is right** **OPEN QUESTION**
lab is gate-validated and limps under the trained policy; legacy is un-gated and looks correct. Whether the gate battery ever scored front-foot symmetry is unestablished. Looking better is not being right.
- 3 Make the eye find moving prey** **READY**
The prey now moves, which it never did during any previous vision work. Add the two-channel split the mouse superior-colliculus literature shows: a wide detector for *something is out there*, a narrow one for *it is 12° left*.
- 4 Put objects in the world** **READY**
Shelter, a warm surface, a threat. The animal is thigmothermic — it takes heat from the ground, not from light, so *bask* is the wrong mechanism as built.
- 5 Add smell** **READY**
The documented prey and predator channel for this species. Real numbers exist: 3.0 → 14.57 tongue-flicks per minute. It can say *food is near*, never *food is there* — no gecko has been shown to localise prey by smell.
- 6 Sleep, memory, learning** **THIN EVIDENCE**
Leopard geckos appear in a 2026 reptile sleep study, so sleep has a starting point. Memory and learning are barely measured in this animal; much would be invented, and this project's rule is that admitted gaps beat invented numbers.

What cannot be built honestly

Recorded so the gaps stay visible instead of being invented later.

- No cost of transport, sprint maximum, endurance or thermal performance curve has ever been measured in *E. macularius*.
- No CTmin for this species or any eublepharid — the cold side of the thermostat has no published floor.
- No field study of the wild animal at all: no home range, activity budget, diet or microhabitat selection.
- No strike distance, speed or capture success for this species. The only feeding kinematics paper filmed at 64 fps and could not resolve the capture phase.
- No visual receptive field, size tuning or velocity tuning for **any** lizard tectum. No prey-selective cell has ever been described in a reptile.
- No cricket walking-vibration spectrum, so a vibration sense has nothing to be calibrated against.
- No tightness-of-fit or crevice-preference experiment — the “tight hide” rule every care sheet states has zero primary support in this species.

The full ledger — 183 hypotheses

#	HYPOTHESIS	VERDICT	EVIDENCE
1	Hind stance height causes the duty failure	Confirmed	compensator: 0.641 → 0.733
2	Limb phase = fore/hind touchdown-lateness difference	Confirmed	$0.320 - 0.123 = 0.197$, exactly the measured error
3	Hind stance height causes the <i>phase</i> failure	Refuted	fixing it moved phase the wrong way, 0.577 → 0.607
4	Fore stance height causes it	Refuted	forelimb table nearly flat; front load fell 0.611 → 0.584
5	Actuator bandwidth is the cause	Refuted	stiffening cut elbow lag $17.0^\circ \rightarrow 2.9^\circ$; phase moved 0.6347 → 0.6283
6	Spine bending supplies the missing stride	Refuted	4× amplitude: +6.5 % excursion, −17 % speed
7	Fore/hind gearing mismatch	Refuted	real, predicts the slip, but equalising it is slower at every step
8	Contact softness lets the feet creep	Refuted	impratio 1 → 100: slip 13.5 → 12 mm. Required $\mu \approx 0.01$ against a floor of 0.9
9	Commanding an earlier touchdown fixes phase	Refuted	commanding 0.197 earlier gave 0.688 — worse
10	Parameter search can reach 6/6	Refuted	~200 generations, four searches; 23.20 → 22.92, tiebreaker only
11	A learned residual can fix the forefoot	Refuted	2×3.01 M steps; best trained 3/6 against base 4/6
12	Trunk pitch lifts the shoulder	Refuted	measured 3.87° against a published 3.72 ± 0.61 — it was correct all along
13	Elbow authority is the missing lever	Confirmed	front load 0.584 → 0.674, unreachable with the elbow locked
14	...and it can be had without cost	Refuted	every passing row had stride CV 0.16–0.38 against a 0.10 ceiling
15	Pressing harder in the base fixes it	Refuted	0.584 → 0.625 then collapse; the press is overwritten during stance
16	Aiming the frozen pose deeper fixes it	Refuted	flat front-only, worse uniform, across −0.6 to −7.0 mm
17	The walking posture rides too high	Confirmed	shoulder 0.125 SVL against a published 0.11; forefoot 3.1 mm airborne

#	HYPOTHESIS	VERDICT	EVIDENCE
18	The forelimb is under-actuated	Confirmed	1 free joint against the hindlimb's 2
19	The limbs are barely articulated	Confirmed	6 of 8 excursions at 0.00–0.37 of published
20	Sprawl can reach the published excursion	Confirmed	55.25° against a published 52.42
21	...without losing the planted foot	Refuted	hind duty 0.631 → 0.094 across the sweep
22	A sprawl-aware stance table rescues it	Refuted	±15° solvable — half what is needed — and 2/6 at zero amplitude
23	The tail was a passive pendulum	Confirmed	every hindlimb excursion moved < 1 % when tail drive was removed
24	A caudofemoralis coupling reproduces the ablation	Confirmed	–20.7 % / –16.5 % against a published –21 % / –17 %
25	One gain reproduces the whole per-joint pattern	Refuted	knee +3.8 % against a published –11 %
26	The food vector is the only privileged channel	Refuted	the controller was steered by the same bearing; both had to go
27	The floor needed a texture	Refuted	it had one — at 0.64 squares per body length. The scale was the fault
28	~70° camera is a guess	Refuted	it makes the 1.6° tracked dot span 1.46 px where 120° gives 0.85
29	The corpus specifies the camera	Refuted	five incompatible recommendations, all INVENTED
30	Cricket escape speed is published	Refuted	absent — only dash length and duration, never divided
31	A walking gecko can catch fleeing prey	Refuted	0.055 m/s against 0.118 m/s; never captured in 40 s
32	A strike can be scored at the control rate	Refuted	a 16–20 ms strike is under one 50 Hz step; 500 Hz needed
33	The no-cheat work landed where it mattered	Refuted	the vision env had its own privileged channel, untouched
34	The eat radius was roughly right	Refuted	0.10 m against a published 4.07 cm strike distance

#	HYPOTHESIS	VERDICT	EVIDENCE
35	Scaling the cheat to zero removes it	Refuted	five slots remain in the observation
36	The camera saw a real food object	Refuted	a green sphere painted on after rendering
37	The brain evidence files are reproducible	Refuted	both pin a body hash no file in the repo has
38	The old drives were roughly right	Refuted	hunger saturated in 67 s where the animal takes days
39	Anchoring the allometry at 40 g is fine	Refuted	inflates metabolism 17 %; measured mass is 19.5 g
40	Hunger measured against the energy reserve	Refuted	1.7 % per inter-meal interval — the animal would never eat
41	Hunger measured in meals matches the animal	Confirmed	70 % at the published 2.33-day interval, unfitted
42	The metabolism and the feeding schedule agree	Confirmed	79 h per meal against a 56 h interval, from separate papers
43	A clipped drive is harmless	Refuted	zero reward for eating when several meals in debt
44	$n = m$ is an acceptable drive exponent	Refuted	drive goes linear; a meal worth the same at any hunger
45	The thermostat can be built now	Refuted	no temperature field exists anywhere
46	Session 5's module was complete	Refuted	fatigue absent; the second channel carried no information
47	The endurance velocity is in m/s	Refuted	gives 15 h at 0.050 m/s against a published >60 min
48	A walking gecko accumulates fatigue	Refuted	0.055 m/s is under the aerobic ceiling — sustainable indefinitely
49	The danger channel drives fear correctly	Refuted	it is mechanosensory, and mechanosensory-alone defence is 0
50	curiosity had no published basis	Refuted	tongue-flick 3.60×, ceiling 0.146/s — right to delete, wrong about why
51	The architecture can be built from the corpus alone	Refuted	the weights are not in it
52	Loop gain 1.0 is acceptable in positive feedback	Refuted	oscillated at zero input, gates cycling 0 → 0.34

#	HYPOTHESIS	VERDICT	EVIDENCE
53	Diffuse drive can be summed regardless of channel count	Refuted	effective gain scales with N
54	Averaging the diffuse drive fixes it	Refuted	cures oscillation, destroys discrimination
55	Self-calibrating the gate is safe	Refuted	collapsed the range; every dopamine level identical
56	Selection is disinhibition, not a maximum	Confirmed	losers are partially released; a WTA cannot do that
57	Sigma-pi salience gates hunting on hunger AND prey	Confirmed	a sated gecko ignores visible prey
58	The GPR parameters are unobtainable	Refuted	Table 5 of arXiv cs/0601004, plus $c = 0.169$ from a CC BY paper
59	The oscillation was an unfixable structural flaw	Refuted	it was the slopes: VL 0.62 keeps the loop gain under one
60	The striatum is driven by cortex	Refuted	Eq 14 drives cortex from thalamus alone; salience goes direct
61	825 switches per 120 s was a model failure	Refuted	50 Hz white-noise salience — the test measured its own noise
62	Distortion can be counted	Refuted	the count saturates; deepening needs a continuous measure
63	Published parameters make the sweep reproduce	Partly	5 of 7 properties; the switching ratio stays inverted
64	Girard 2005 Table 5 holds the GPR parameters	Refuted	it is a re-tuned robotic variant; six independent sources give 0.9 / 0.3, not 0.8 / 0.4
65	The oscillation at zero salience had one cause	Refuted	it had three, and each hid the next
66	The caller's control rate can be the solver's rate	Refuted	at 50 Hz the resting output oscillates over 0.143–0.205; at 200 Hz it is exact
67	The diffuse STN drive caused the instability	Refuted	the drive is published and deliberate; the integration step was the fault
68	A guard pinned to one publisher's name is a good guard	Refuted	it failed when the citation was corrected, which is backwards

#	HYPOTHESIS	VERDICT	EVIDENCE
69	The sweep evidence was reproducible	Refuted	generated by a throwaway script; could not be re-run when parameters changed
70	Canonical parameters make the sweep reproduce	Partly	6 of 7; switching ratio improved $0.3\times \rightarrow 0.55\times$, still inverted
71	The open-loop CPG contributes ~2% of the walking	Refuted	that figure is the legacy profile on the wrong body. The accepted walker is the open-loop base: 19 strides, 1.1892 Hz measured against 1.1888 commanded, duty 0.733
72	The brainstem's declared output has somewhere to land	Refuted	the cord is <code>(t.freq + offset) % 1.0</code> — closed form, no phase state, no drive input. 1 of 4 drive channels is consumable
73	The switching-bout definition needs the supplementary materials	Refuted	it is in the main text, §3.1.3, freely readable — a wall recorded without being verified, for the second time
74	The $0.55\times$ switching ratio is a failed reproduction	Refuted	the published number counts bouts over two named task sequences against a task-topology minimum of 7, and the authors explicitly declined fixed-interval counting — which is what our tool does
75	A published <i>disembodied</i> acceptance test exists	Confirmed	Prescott 2024 Study 1 workbook: 61 dopamine levels $\times$ five selection classes, CC BY, obtained
76	The module reproduces that table	Partly	every ordering reproduces and distortion onset lands on the published step; partial selection 17–47 points high, clean 15–31 low
77	The module computes the paper's distortion	Refuted	ours divides losers by the winner; Eq. 3 divides by the total and doubles it
78	The registry holds a published stride frequency	Refuted	116 entries, and the only frequency is the invented 1.1888 Hz lock
79	This species changes gait with speed	Refuted	relative phase not significant over 0.2–1.1 m/s; $44\pm 1.1\%$ walking against $43\pm 1.8\%$ running
80	The published EMG pattern is wired into the controller	Refuted	<code>common/emg_pattern.py</code> is imported by tests and nothing else
81	MLR stimulation has never been done in a gecko	Refuted	walking, phonation and left/right turns elicited from the midbrain of Gekko gecko — the one study in the target family
82	The scout's settle loop converged (<i>a scout's own method error</i>)	Refuted	breaking on two equal gate vectors fired during the clipped-at-zero transient and reported 100% no-selection at every level

#	HYPOTHESIS	VERDICT	EVIDENCE
83	The scout's first walker measurement used the accepted body (<i>same</i>)	Refuted	it used the default body and legacy profile; the accepted body is identified only by a hash inside a gate evidence file
84	The winner failing to saturate was a defect in our model	Refuted	it was a model mismatch; every constant we had was right for a different model
85	The 2024 paper uses the 2001 ($1 \pm \lambda$) dopamine mechanism	Refuted	dopamine modulates the striatal output slope about a pivot, not the input gain. The authors call it "new DA" in their own source
86	The striatal threshold is 0.2	Refuted	0.1 in the basic variant, 0.15 in the extended
87	Salience reaches the striatum directly	Refuted	through a leaky cortical relay — and the extended model splits it 0.5/0.5 with the thalamic return
88	The supplementary materials were unobtainable	Refuted	on disk: the authors' C++ source and the Figure 5 data workbook
89	Figure 5 came from the basic variant	Refuted	the shipped harness sets <code>EXTENDED 0</code> , but the basic variant peaks at 8% clean selection where the extended reaches 79%
90	The published distortion carries the paper's factor of two	Refuted	the authors' code has no factor of two, and the published range 0–0.217 fits the unfactored form
91	A faithful implementation reproduces the published table	Partly	akinesia exact, peak clean 79.9 against 78.6, efficiency 0.996 against 0.999 — but the curve sits +0.10 higher in dopamine
92	The residual disagreement is a shape error	Refuted	axis fit slope 1.08 — a constant shift, not a scaling or a deformation
93	The sampling behind Figure 5 is recoverable	Refuted	percentages exact to 1e-5 imply 100,000 competitions per level; the shipped harness sweeps 202 ramp points
94	The winner-take-all comparison arm reported its own result	Refuted	<code>gates()</code> returned all zeros one line after <code>step()</code> named a winner
95	The environment has a behaviour arbiter to replace	Refuted	it has none. It is reward-driven, and the <code>target_interest</code> the plan named lives in the superseded drive module
96	The selector can be given control of the animal now	Refuted	only one behaviour has a controller. A six-way chooser on a one-behaviour body would look like integration and mean nothing

#	HYPOTHESIS	VERDICT	EVIDENCE
97	Persistence needs an invented stickiness constant	Refuted	the published cortico-thalamic loop supplies it structurally — the invented gain of 0.15 is deleted, not re-tuned
98	The two brain modules can run together on live data	Confirmed	hypothalamus → selector every control step inside the simulation; a tired gecko rests, and a rested hungry one explores
99	The plan's "bit-identical at zero coupling" test is achievable	Refuted	an accumulated phase and a closed-form one differ in the last bits, and the clock snaps to exact cycle boundaries an accumulator never lands on
100	An integrated oscillator can reproduce the accepted walker's rhythm	Confirmed	max deviation 3.3×10^{-11} of a stride over 60 s of walking
101	The foot order is the obvious one	Refuted	it is HL, FL, HR, FR — restating it would have planted a silent index mismatch
102	A single-step check shows the oscillator and the clock differ (<i>my own test error</i>)	Refuted	they agree exactly for the first few steps; divergence needs accumulation, so the test proved nothing
103	The dopamine discrepancy is a constant offset	Refuted	neither an offset nor a scaling — the mechanism was wrong
104	The published model modulates the striatal output slope	Refuted	it multiplies the afferent input: $(1 + \lambda)$ to D1, $(1 - \lambda)$ to D2, threshold 0.2, quoted verbatim in Supplementary Methods §5
105	Reading the files an archive contains is enough	Refuted	the builders every program includes are absent; the two present are the branch every program disables
106	Session 6e's "the whole model was wrong" was right	Partly	right about the structure. Wrong about dopamine and the threshold, which the first implementation had right
107	The published sweep is memoryless	Refuted	it is hysteretic; cold-starting each cell moves the table by ~1 percentage point
108	The corrected model reproduces the published table	Confirmed	mean error 0.103 pp, all four onsets exact, peak at 0.22 = published, axis fit slope 1.000, intercept 0.000
109	Attaching the integrated cord changes the gait	Refuted	every gate identical; hind duty 0.733418 both; speed differs by 3.2×10^{-14} m/s
110	The parity harness is the official gate scorer	Refuted	it records at its own rate, so only the comparison is claimed, never the absolute count

#	HYPOTHESIS	VERDICT	EVIDENCE
111	isinstance is safe for path-loaded modules	Refuted	the same file under two names gives two classes, and a perfectly good cord was rejected
112	The food detector can see the food	Refuted	it tested for green; the prey that replaced the painted marker is brown. Returns exactly 0.000 at every illumination from 0.4× to 3.0×
113	The no-cheat world is in use	Refuted	referenced only by its own test and its generator. No training or evidence run has ever loaded it
114	The old detector at least worked for its own marker	Refuted	absolute channel thresholds, so the painted sphere also scored 0.0 at gain 0.4 — it failed on illumination change too
115	A colour detector can find the prey in this world	Partly	it can, but the margin is thin: the nearest confuser is the gecko's own spots at 0.145. A brown cricket on sand is genuinely camouflaged
116	The policy receives an image	Refuted	the encoder average-pools to 128 numbers over the whole visual field — "how much", never "where". There is no retinotopic map at all
117	A rectilinear camera samples uniformly, as the animal does	Refuted	corners sample 4× finer than the centre at 120° — an inverse fovea, worst exactly where the animal aims
118	The default build can resolve what the animal tracks	Refuted	the published 1.6° tracked dot spans 0.516 px at fovy 120 — not resolvable
119	The eye's field of view is sourceable	Refuted	still NOT_IN_CORPUS for this species; the five corpus recommendations remain invented
120	Leopard geckos have a tapetum	Refuted	unverifiable from any primary source — every confident claim traced to a pet-care site. Do not build one
121	A gradient flow estimator on the cell grid is accurate	Refuted	2.07x too large at every velocity — the drum's stripes span four cells, where a central difference underestimates the gradient by a third
122	Estimating at pixel resolution fixes it	Partly	2.07x to 1.15x. The residual is the inverse fovea: pixels are not uniform in angle
123	Differencing against azimuth removes the bias	Confirmed	ratio 1.02 against truth across 10-40 deg/s
124	The published monocular asymmetry can be reproduced	Confirmed	naso-temporal gain exactly 0.000 at all three velocities — a symmetric estimator cannot do this
125	The binocular-versus-monocular comparison can be run	Refuted	the body has ONE camera. There is no second eye to cover. Recorded as blocked, not as passed
126	The optokinetic gain should approach 1	Refuted	published gain is 0.9 and falls to 0.7-0.8 by 40 deg/s. Perfect stabilisation would be over-reproduction

#	HYPOTHESIS	VERDICT	EVIDENCE
127	The tectum finds prey on synthetic stimuli	Confirmed	correct bearing at every position, silent on a still world, silent on pure self-motion
128	It finds prey in the environment	Refuted	fires on 99.3% of frames and the bearing correlates -0.03 with the prey's. Mean error 63°
129	The failure is the map being too coarse	Partly	full pixel resolution reaches only $+0.10$. Resolution is part of the cause, not the whole
130	The prey is resolvable in the map at working range	Refuted	it covers 0.69 of one cell at 0.30 m and 0.41 at 0.50 m
131	My self-motion control demonstrated what I claimed	Refuted	the size test was doing the work; field subtraction is worth a fourfold reduction, not elimination
132	The order of the two rejection tests is arbitrary	Refuted	size test after subtraction leaks 0.066; before, 0.000
133	Subtracting the surround stops the tectum hunting its own optic flow	Refuted	it subtracted a scalar, and <code>argmax(clip(x-c,0,None)) == argmax(x)</code> for any scalar <code>c</code> . Identical on 20000/20000 random maps — it had never changed a single reported bearing. Replaced with a spatially varying local surround, which moves the answer on 1735/2000
134	A sharper eye finds the cricket	Refuted	four configurations, render 64→512 px and receptors 64→256. Correlation with true prey bearing ≈ 0 in all four. I had told the user resolution was the fix; it is not
135	There is prey motion for a motion detector to find	Refuted	<code>prey_total_travel_mm = 0.0</code> across 1800 frames. <code>FleeingPrey.step</code> only moves within 0.075 m while detection range is 0.30–0.50 m. The prey is stationary
136	The prey is in the camera's field of view	Refuted	present on 18.7% of frames. Off the rendered image four frames in five
137	An open-source retina model can be adopted	Refuted	the only candidate, <code>openretina</code> , is a mouse retina (MIT code, CC-BY-SA weights) and retraining needs gecko recordings that do not exist for any reptile
138	Geckos head-bob to judge prey distance by motion parallax	Refuted	no gecko study shows it, and none shows it for prey in any lizard. It is a stated hypothesis in other lizards, never demonstrated. Every confident claim traced to pet-care sites. I asserted this to the user as fact. The one lizard-clade distance mechanism that IS established is chameleon accommodation, which is not motion-based
139	Stationary prey is undetectable by a visual predator	Refuted	mice capture fresh-frozen crickets as fast as live ones — 8.06 ± 1.08 s vs 7.6 ± 1.34 s, $n=8$, n.s. Motion is not required. The blindness is our architecture's, not nature's
140	A better tectum would find still prey	Refuted	in frogs the pretectum handles stationary objects and the tectum handles motion: pretectal lesions leave feeding and threat avoidance intact but the animal walks into a barrier it can see. A tectum-only model is correctly blind to a frozen cricket. This needs a second structure, not a better one

#	HYPOTHESIS	VERDICT	EVIDENCE
141	One prey channel is enough	Refuted	mouse superior colliculus splits it: wide-field (RF 700–900 deg ² , not direction-selective) triggers approach from >22 cm; narrow-field (<200 deg ² , direction-selective) supplies bearing during pursuit. Suppress the second and the animal reaches the prey and stalls beside it. Echoed in zebrafish posterior/anterior tectum
142	Prey selectivity is a conjunction of features	Refuted	the one worked-out vertebrate is the opposite. The toad tectum's default is attack anything that moves; selectivity is a subtractive pretectal veto. Lesion the pretectum and the toad attacks everything moving. Our tectum ANDs its criteria — architecturally backwards
143	Elevation in the visual field is a soft weight	Refuted	it is categorical. Identical looming disc: 75% escape overhead vs 53% and much slower at the side. Identical small sweeping disc: freeze overhead, 80% approach at the side. Floor-mounted looms produced zero escapes
144	The prey velocity band is a free parameter	Refuted	four independent lineages converge on the same order: toad behaviour peaks 30–60 °/s and dies above ~200; pit viper tectum (a squamate) is near-silent at 5 °/s and peaks ~50; mouse looming escape needs 35 °/s. This is the most robust cross-species number found
145	Our gecko basks	Refuted	it is thigmothermic. Body temperature tracks the substrate at $r^2=0.97$ against air at 0.92, $n=12$. It takes heat by lying on warm ground, not from light. And preferred temperature is not a set point but a diel ramp, ~26 °C morning to ~30 °C at dusk — the rise is proposed as what triggers evening emergence
146	A visual threat channel models this animal's escape	Refuted	for this species, $n=42$: defensive reaction probability is 0.21 to snake smell and 0.07 to the sight of a snake. Factorially: chemical $\chi^2=8.098$, $p<0.0044$; visual $\chi^2<0.01$, $p>0.9$. Verbatim: "the initiation of costly defensive action remains strictly gated by chemoreception"
147	Smell can localise prey	Refuted as unsupported	tongue-flicking is real and quantified for this species — 3.0/min baseline → 14.57/min to cricket chemicals, $n=7$ — but the assay presents the swab 1 cm from the snout, in plain view. It demonstrates discrimination, never localisation. Cooper flagged the missing experiment in 1998 and nobody has done it since
148	<code>cost_of_transport = 0.73</code> is a leopard gecko measurement	Refuted	it is <i>Teratoscincus przewalskii</i> , a different family. <code>docs/research/gecko_scorecard.md</code> had it right, tagged <code>[TP]</code> and defined as a proxy for energetics only; <code>config/proxies.yaml</code> — the file the code reads — said <code>species: E_macularius</code> , <code>confidence: verified</code> . Corrected. No cost of transport, sprint maximum, endurance or thermal performance curve has ever been measured in this species
149	456 tests pass	Refuted	8 <code>test_session2_gate.GateContracts</code> tests fail, and they fail in a clean worktree at commit e4196d8 — they are not caused by session 8d's edits. See below
150	No feeding kinematics exist for this species	Refuted — my own overstatement	Delheusy, Brillet & Bels 1995, <i>Amphibia-Reptilia</i> 16(2):185–201, $n=6$ adult males, 64 fps, 22 mm crickets, full Table 1. I turned one agent's "could not retrieve the numbers" into "does not exist". What is absent is narrower: the capture phase was unquantifiable at 64 fps (5 usable cycles from 2 animals) and is excluded from the paper's tables
151	The gecko takes prey with its tongue	Refuted	jaws alone — the standard scleroglossan pattern, no tongue prehension. And the orienting rule is published: the head is aligned transverse to the prey's long axis. Capture is a single open–close cycle with no slow/fast subdivision: ~80 ms, peak gape ~37° at 47 ms, head translating ~8 mm horizontally and ~28 mm vertically
152	Eyelid geckos cannot lick their own eyes	Refuted	Delheusy et al. 1995, verbatim: "La langue se déplace hors de la cavité buccale et atteint parfois l'oeil" — the tongue sometimes reaches the eye, during post-feeding licking and grooming. The "cannot" claim traces only to care sites. Separately: labial-licking is a different behaviour and the words eye, ocular, eyelid appear nowhere in the paper that defines it

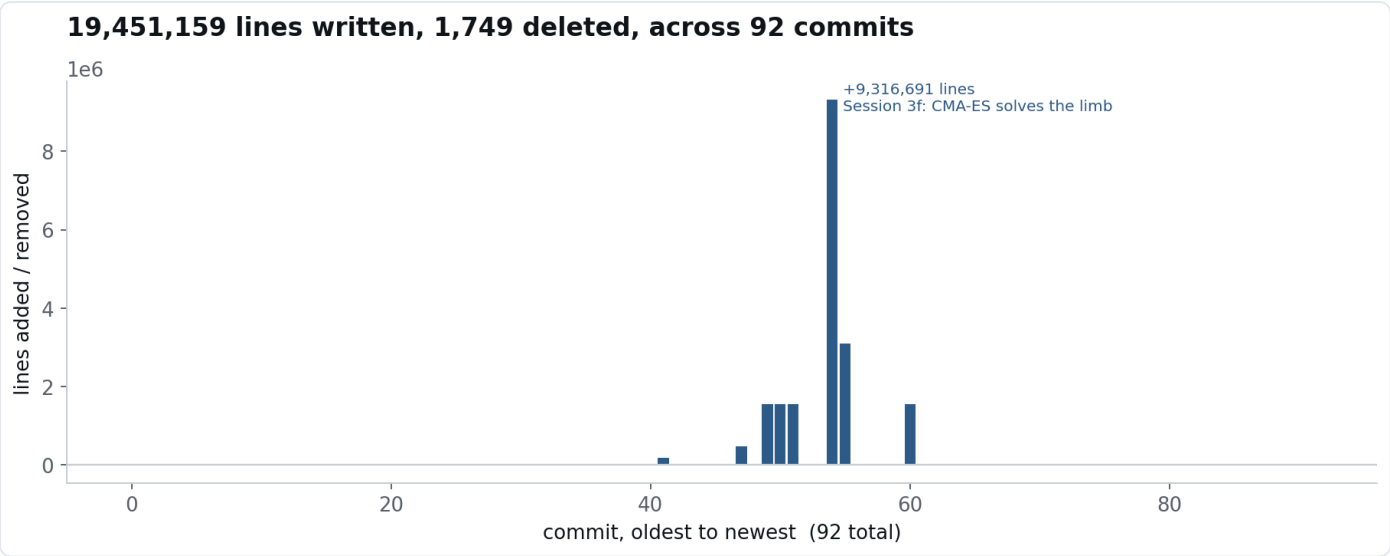
#	HYPOTHESIS	VERDICT	EVIDENCE
153	This animal navigates by what things look like	Refuted	given a conflict between geometry and visual features, <i>E. macularius</i> preferentially used geometry (Kundey 2021, Behav Processes 188:104412). Its world model is shaped more by layout than by appearance — which is the opposite of what a vision-first build assumes
154	The circulating leopard-gecko strike numbers have some source	Refuted, source pinned	"486 m/s ² , 5.8 m/s, 35 cm" is Wainwright, Kraklau & Bennett 1991, J Exp Biol 159:109–133, tongue projection in <i>Chamaeleo oustaleti</i> , n=3 sequences from 2 animals. A leopard gecko does not project its tongue 35 cm
155	Sleep is unmeasurable in this animal	Refuted	<i>E. macularius</i> is one of seven lizard species in Bergel et al. 2026, Nat Neurosci 29:543–550 — sleep-dependent infraslow rhythms conserved across reptiles and mammals, EOG under each eyelid. It closes its eyelids to sleep; the tokay cannot. Brain 5–8 has a starting point that did not exist when they were listed as ✗
156	The gecko vibration band is press-release-only and unusable	Refuted	read from the paper itself. Han & Carr 2024, n=31 tokay geckos, 38 single units: saccule → nucleus vestibularis ovalis → torus semicircularis, a channel entirely separate from hearing. Best frequencies 50–200 Hz, mode ~100 Hz; threshold –53 to –23 dB re 1 m/s ² , mean –42.6; phase locking VS 0.89. Sound at 15 dB above the loudest vibration used evoked nothing — it is vibration, not sound
157	The ear covers low frequencies, so vibration adds little	Refuted	no lizard auditory-nerve fibre has ever been recorded below ~200 Hz CF, across four primary datasets, and the audiogram is a flat insensitive shelf from 100–500 Hz. The vibration channel covers exactly the band hearing cannot reach. They are not redundant; they are complementary
158	No lizard has been shown to find prey without seeing it	Refuted	<i>Scincus scincus</i> , buried in sand, detects crickets moving on the surface up to ~15 cm away, extracts direction, emerges and captures. It responds far less to dead insects at the same distance, so it is vibration and not smell, and it discriminates prey speed — faster prey, faster emergence. It also plunges its head into the sand, and experiments show this is what couples it to the substrate
159	No usable audiogram exists for this animal	Refuted, with a substitution	none is accessible for <i>E. macularius</i> — Werner et al. 1998 and 2008 hold it and both are hard-paywalled with no open copy anywhere. Two usable substitutes: eight gekkotan species best 22–72 dB SPL at 0.5–1.0 kHz (Werner et al. 2008), and a free, full audiogram for <i>Coleonyx variegatus</i> , an actual eublepharid — best ~19 dB SPL at 500 Hz at 21 °C (Wever et al. 1964, PMC300117)
160	One audiogram describes this animal	Refuted	gecko hearing is temperature-dependent in both sensitivity and best frequency: as the animal cools it hears less and its best frequency shifts downward. 249 sensitivity functions, 50 animals, 15–40 °C. A fixed audiogram is biologically wrong for an ectotherm, and this project's animal already has a body temperature
161	Vibration prey-detection is demonstrated in geckos	Refuted	it has never been tested behaviourally in any gecko. Han & Carr say so themselves — "no behavioral role for a dedicated vibration receptor has been demonstrated" — and their one mention of predator-prey sits bracketed with wind and rain. I first recorded this as a mechanistic coincidence worth noticing — cricket signals below 120 Hz, saccular channel 50–200 Hz, "the bands line up". The next search refuted that too. See #162–164: on the numbers the gecko is not sensitive enough and is tuned to the wrong band. The tympanic ear being ~55 dB down at 100 Hz stands
162	A gecko could feel a cricket walking	Refuted on the numbers	measured arthropod footsteps on sand are 4x10 ⁻⁴ to 1.5x10 ⁻³ m/s ² (Devetak et al. 2007, four species). The most sensitive gecko saccular cell ever recorded thresholds at –53 dB re 1 m/s ² = 2.2x10 ⁻³ m/s ² , mean 7.4x10 ⁻³ . The loudest measured footstep sits below the best single cell's threshold, before any distance damping. A scorpion does this because its receptors trigger at ~1 nm (10 A) - the gecko's need roughly 19 nm. Twenty-fold short

#	HYPOTHESIS	VERDICT	EVIDENCE
163	The gecko's vibration band is the right band for sand	Refuted	sand is a lossy low-pass medium: Rayleigh waves at 40-50 m/s dry, damped 0.26-2.61 dB/cm at 300 Hz, worse as grains get finer. The band that actually propagates usefully is 200-500 Hz (Brownell & van Hemmen 2001). The gecko's saccular channel is 50-200 Hz - it sits below the window where sand carries prey information. And a cricket's own low-frequency signal peaks at 43 Hz, below both
164	Detection range is a property of the animal	Refuted	it is a property of the ground. Antlion mean reaction distance is 3.3 cm in fine sand (≤ 0.23 mm grains) and 12.3 cm in coarse (1-1.54 mm) - a near fourfold change from substrate alone, in the same species. Any vibration range this project ever quotes is meaningless without the grain size beside it
165	The privileged target channel is removed everywhere	Refuted, third occurrence	<code>GeckoBrainEnv</code> built the walker without naming <code>privileged_target</code> , so the walker's own <code>True</code> default won and nothing in the brain env could override it. Five numbers - exact egocentric direction, distance and bearing to the goal, straight out of the physics engine, noiseless, at any range. All 12 checkpoints in the repository record <code>proprio_dim 92</code> , including the ones described as pure-vision. See #26 and #33 for the first two times this was declared removed
166	It is present but the policy ignores it	Refuted, and this is the expensive one	<code>tools/oracle_ablation.py</code> zeroes the five slots at inference, keeping the vector 92 long so the checkpoint still loads. The walker moves further with them zeroed (0.0857 $\rightarrow$ 0.1103 m) and progress toward the goal collapses 0.0664 $\rightarrow$ 0.0060 m, a 91% loss. The legs are fine; the navigation was the oracle. Retiring it needs a retrained walker, so the flag is exposed and declared rather than flipped, and <code>info["walker_oracle"]</code> now reports it every step
167	The sessions 3-4 gait evidence is unverifiable	REFUTED, and the claim was mine	I reported that the recorded body hash <code>756765356417749c</code> matched no committed version and that the accepted walker's evidence therefore pointed at a body not in the repository. The physics was identical all along. The comment-free ElementTree serialization of the current body is <code>b178bf26b644b1d3</code> , which is exactly what <code>artifacts/evidence/morphology_reproducibility_line_endings_20260905.md</code> recorded for that file. Both committed versions in git history share that same physics hash. A comment changed, not the animal. The evidence was always valid
168	Eight gate contracts are failing	Refuted -- it was one	the body check raised before any other contract could reach its own assertions, so seven unrelated tests reported the wrong failure for four sessions. Fixing the one check turned all eight green. <code>common/body_identity.py</code> now compares what MuJoCo reads rather than what the bytes say, and the evidence writer records both fingerprints, because recording only the raw one is what made this unverifiable in the first place
169	Adding a strike makes hunting possible	Refuted	necessary, not sufficient. The prey bolted at a FIXED 0.075 m regardless of how it was approached, the strike must launch from 0.0203 m, and the prey escapes at 0.118 m/s against a 0.055 m/s walk. Measured over 60 s of pursuit: closest approach 51.1 mm, zero strikes. The gecko could not physically reach striking distance by any route
170	A stalk is just a slower walk	Refuted	it is only a behaviour if the PREY can tell the difference. The published ethogram names <code>walk slow motion</code> , "walking with a strongly reduced speed, mostly in context of prey capture" -- meaningless against a prey with a fixed flee radius. The flee radius now scales with approach speed: 75 mm at a charge, 12 mm at a creep, which is inside strike range. Direction supported (flight-initiation distance rises with approach speed; crickets sense predators through cercal air-current receptors); magnitude INVENTED and the capture rate rests on it
171	The predator's own speed is what frightens the prey	Refuted	the predator point is the gecko's NOSE, which sways with the gait: measured 0.129 m/s of apparent approach during a creep that netted almost nothing. Reading gait sway as a charge made stalking impossible. What a prey animal has to go on is how fast the threat is getting CLOSER, so closing speed is measured instead, floored at zero

#	HYPOTHESIS	VERDICT	EVIDENCE
172	Striking distance can be measured in three dimensions	Refuted	the nose sits 13.7 mm above a prey that is 9 mm off the floor, so the vertical gap alone eats most of a 20.3 mm trigger. Closing it is what the strike is FOR: Delheusy et al. 1995 measured the head translating ~8 mm horizontally and ~28 mm vertically in the 80 ms cycle. The trigger is horizontal
173	The prey can own capture once a strike exists	Refuted	proximity capture respawned the prey the instant it entered striking range, so the strike could never fire -- measured 0 strikes across 1050 frames with the prey placed in range on every single step. Capture belongs to the strike or to proximity, never both
174	The frozen walker can deliver the animal to striking distance	Refuted, and this is the live blocker	it cannot. It arrives within 4 cm of a goal and its nose trails its trunk by about 5 cm, so "the animal arrived" and "the mouth is on the prey" are five centimetres apart. With prey motion, a working stalk, a collapsed flee radius and a target aimed past the animal, the best approach across 700 steps was 34.5 mm against the 20.3 mm needed, oscillating rather than converging. The walker was trained to reach a goal, not to place a mouth on a 9 mm cricket
175	The strike itself works	Confirmed	with the prey placed in range -- which the approach cannot do -- 78 strikes, 63 caught, 14 missed: 81 % against the published 82.9 %, unfitted. <code>artifacts/video/hunt_session9.mp4</code> part 2
176	The Session 9 videos show a changed gait	Refuted -- they show HALF SPEED	every Session 9 video tool appends one frame per 0.02 s brain step and wrote them at 25 fps. That plays 20.0 s of simulation over 40.0 s of video: 2.0x slow motion, in all four clips. The Session 4 renders were 50 fps and real-time, which is why they looked right. The "lag" the user saw was a container header. Gait, body, checkpoint and physics unchanged -- 481 tests, three independent body-hash confirmations, walker speed 0.054 vs 0.055 m/s. Fixed to 50 fps; <code>accepted_walk_replay.mp4</code> was already real-time because it samples the trace at the output rate rather than per step
177	The brain runs the accepted walker	Refuted	the 4/6 gate evidence was measured under <code>gait_profile="lab"</code> (<code>artifacts/evidence/lab_frozen/report.json</code>). <code>GeckoBrainEnv</code> never passes the parameter, so <code>GeckoWalkEnv</code> 's <code>"legacy"</code> default wins. Same body, same frozen checkpoint -- model sha256 <code>77b7a99d...</code> is byte-identical to the one the evidence used -- but a different CPG scaffolding underneath. Same shape of defect as #165: a default won because nobody named the parameter. Everything built in Session 9 sits on the un-gated profile
178	The Session 4 renders can be regenerated	Refuted	the script that made <code>renders/session4/.mp4</code> was never committed. Only <code>utils/render_trace_video.py</code> is in the history, and it cannot produce them: it replays a trace against the archived BODY file, so it draws the default floor, has no prey, and disables shadows. The setup is reproducible; the file is not
179	The accepted walker can be re-run live	Refuted RETRACTED same session -- it was my harness	loading the accepted checkpoint into a live <code>gait_profile="lab"</code> env makes it fall after 1.2 s, in both the body XML and the world XML, with and without a distant target. Cause located but not fixed: <code>lab_controller_snapshot</code> is None in the live env, so the lab base controller is not configured -- the evidence protocol carries a whole reward-calibration and controller block that a bare constructor does not reproduce. Found within the hour: <code>GeckoWalkEnv</code> defaults to <code>control_mode="raw"</code> , which builds no CPG at all, and the frozen policy is a RESIDUAL that rides on one. With <code>control_mode="cpg_residual"</code> the accepted walker runs the full 20 s with zero falls, 1.642 m of path at 0.082 m/s, and <code>lab_controller_snapshot</code> is populated. The walker was never broken; I gave it no scaffolding to stand on. Recorded rather than deleted, because I reported it to the user as a finding before checking my own harness

#	HYPOTHESIS	VERDICT	EVIDENCE
180	Running the brain on the gated profile costs something	Refuted -- it costs almost nothing	<code>gait_profile</code> is now a real parameter on <code>GeckoBrainEnv</code> and reaches the walker. Measured over 500 brain steps, same seed, same action: legacy 1.041 m (0.1041 m/s), lab 1.021 m (0.1021 m/s), zero terminations either way -- a 2 % difference. <code>lab_controller_snapshot</code> is populated under lab and None under legacy, which is the honest check that the profile actually took effect rather than merely being stored. The default stays <code>legacy</code> because every brain checkpoint was trained against it and flipping a default quietly is how this class of defect gets made; <code>info["gait_profile"]</code> and <code>info["gate_validated_profile"]</code> now report the truth every step
181	The camera shake is the ground sliding	Refuted twice, and the second answer is the right one	first I blamed the sliding ground and bolted the camera to the world, which stopped the shake and let the animal walk out of frame. Then I blamed the azimuth swinging with heading, which was real but not the whole of it. The remaining shake is the camera pointing at the trunk, and the trunk rises, falls and yaws once per stride -- so the whole world jitters at stride frequency. Following a smoothed path at fixed height instead: measured background jitter 1.538 -> 0.000. Two further filming faults fell out of it: a 1 s smoothing constant left the camera ~10 cm behind a 0.0975 m/s animal and framed it from the rear, and a rear-quarter azimuth makes a sprawling lizard read as flat and splayed while it is fully upright -- up-vector 0.998, never below 0.99, trunk 19-27 mm off the floor for all 1000 frames
182	The push-then-drag is in the Session 4 walk too	Refuted -- I filmed the wrong profile	the user said their demo does not drag, and they were right. Front-foot duty factor against a 0.70 target: lab FL 0.203 / FR 0.591 -- asymmetric by nearly threefold, one front foot loaded a fifth of the step and the other over half, which IS the limp. legacy FL 0.483 / FR 0.500 -- balanced, and it reads as a walk. Every A/B render I made this session used <code>lab</code>; the Session 4 demo is <code>legacy</code>. So #9j's pixel-identical result stands as a statement about the CODE and answers a question nobody asked. The uncomfortable part: <code>lab</code> is the gate-validated profile and it is the one that limps. Whether the gate battery scored front-foot symmetry at all, or scored it and accepted the failure as one of its two unmet checks, is the next thing to establish -- looking better is not the same as being right
183	The limp is the profile	Refuted -- it is the POLICY, and the ledger already said so	#182 blamed <code>lab</code> versus <code>legacy</code>. Wrong again. With the residual switched OFF -- the hand-written base, no policy -- both profiles balance: lab FL 0.432 / FR 0.454, legacy FL 0.463 / FR 0.491. The asymmetry only appears when <code>models/v4_5b_speed_polish_1m</code> is loaded on top. The accepted walker IS the open-loop base -- #71 states it outright, #11 measured the trained residual at 3/6 against the base's 4/6, and <code>lab_zero_residual/report.json</code> records controller "zero residual with contact reflex" in its own protocol field. I read all three this session and still loaded the checkpoint into every clip. So every video I sent was the REJECTED walker, and I explained its limp away three times -- as a camera artefact, a frame-rate artefact, and then as an under-actuated forelimb the user would have to accept. The forelimb IS under-actuated (#18) and that is real; it is not what made the animal limp on screen

Every commit — 92



HASH	DATE	SUBJECT	FILES	+	-
11f6cf7	2026-06-08	Initial commit	1	+201	-0
53face5	2026-06-08	Initial NeuroGecko code release	24	+2,330	-0
bc91977	2026-06-08	Merge branch 'main' of https://github.com/12ziyad/NeuroGecko	1	+201	-0
481a226	2026-06-09	Add V4.2.2 front-foot correction and dual-view rendering	5	+163	-25
e1404c6	2026-06-09	Add V4.2.3 front-loading controller	2	+123	-96
c486d90	2026-06-09	Add V4.2.4 escape-proof front-load gate	2	+68	-57
152fefc	2026-06-09	Add V4.2.6 front-load polish and knee residual cap	2	+6	-4
6770779	2026-06-10	Add V4.2.7 front duty reward gate	1	+51	-1
e522bb5	2026-06-11	Add V4.2.8 contact tracking reflex curriculum	4	+95	-8
eff902e	2026-06-11	Tune V4.2.9 reflex wean controller	1	+2	-2
fdccbc6	2026-06-12	Add V4.3 velocity tracking experiment	2	+56	-1
570bd01	2026-06-12	Add V4.5 matched stance cadence experiment	3	+22	-15
d2e845b	2026-06-12	Add V4.6 front de-stick experiment	2	+4	-4

HASH	DATE	SUBJECT	FILES	+	-
0f6c7ca	2026-06-12	Add V4.5 final locomotion base	2	+4	-4
0e7858c	2026-06-12	Record V4 locomotion final results	1	+37	-0
1c6e2ef	2026-06-12	Accept V4.5B as final locomotion champion	1	+20	-9
aad003a	2026-06-13	starting brain	5	+693	-0
cb93609	2026-06-13	Add AGWM brain environment wiring	2	+29	-7
9541f8f	2026-06-13	Add trainable visual brain policy	7	+424	-1
f72ca1c	2026-06-13	Add brain curriculum and stable viewer controls	4	+244	-17
2b5c158	2026-06-13	Add privileged food taper curriculum	3	+165	-7
d0f9deb	2026-06-13	Make brain camera see food and add privileged dropout curriculum	4	+235	-7
d60f33f	2026-06-14	Use mouth distance and add brain resume support	3	+139	-27
a505ade	2026-06-14	Fix brain food reward scaling	3	+22	-5
cd014fe	2026-06-15	Add brain watcher stochastic diagnostics	1	+59	-3
342eaa1	2026-06-15	Rebalance brain food reward	1	+11	-7
0a57988	2026-06-16	Add brain visual lock reward and checkpoints	3	+55	-21
3460cb0	2026-06-16	Use visual delta reward and fix brain checkpoints	3	+34	-8
7e90df7	2026-06-16	Add brain food cue radius curriculum	3	+97	-9
6953dcf	2026-06-17	Add oracle action eval mode	1	+10	-1
11d0a08	2026-06-17	Add recurrent PPO brain support	2	+130	-20
a3de4ee	2026-06-18	Add privileged oracle behavior cloning	4	+485	-7
9efac13	2026-06-18	Fix BC diagnostics timing	2	+4	-2
c8c616e	2026-06-18	Add BC policy debug script	1	+396	-0
b370892	2026-06-18	Add Patch37B DAgger recovery BC	2	+366	-10

HASH	DATE	SUBJECT	FILES	+	-
407e5d5	2026-06-18	Add Patch38A visual distillation preflight	5	+1,117	-6
c05ca81	2026-06-18	Add Patch38C visual DAgger recovery	4	+823	-15
2681c9e	2026-06-20	Add Patch39A visual search baseline	3	+703	-0
ddeb105	2026-06-20	Add Patch40A drive-based GeckoBrain controller	2	+961	-0
44a921c	2026-09-05	Recover and verify the cloud model lineage before implementation	6	+1,308	-1
0c9f639	2026-09-05	Add measured baselines, camera isolation and verified checkpoint recovery	25	+3,966	-24
9960117	2026-09-05	Add constrained lab morphology, shared gait calibration and verified recovery evidence	41	+184,185	-63
3efa2f6	2026-09-05	Verify candidate motion and cross-platform reproducibility without altering legacy models	15	+26,271	-19
a7517e6	2026-09-05	Step 0: record true 250 Hz physics and report failed entrainment honestly	7	+16,578	-13
faa5dda	2026-09-05	Step 1: charge lab effort on total actuator work while preserving legacy reward	5	+276	-8
992b062	2026-09-05	Step 2: preserve force and collision-geometry diagnosis before controller edits	7	+33,232	-0
790edd4	2026-09-05	Step 2: verify historical command reconstruction with 19 force-only tests	3	+247	-1
ffc7be5	2026-09-05	Step 3: improve and verify the lab base; stop training on failed support and timing gates	80	+463,881	-25
ad54e9a	2026-09-05	Record Session 3 geometry baseline and hypothesis protocol	3	+9,930	-0
9eee048	2026-09-05	Session 3: implement, fix and test hind stance compensation; Gate 2 still fails	8	+1,558,377	-3
8153e8b	2026-09-05	Session 3b: generalise stance compensation to all four limbs; Gate 2 now 4/6	11	+1,554,407	-42
552c91b	2026-09-05	Session 3c: limb phase is late touchdown; stride length is geometrically unreachable by the limbs	4	+1,552,508	-0
8c67937	2026-09-05	Session 3d: stride length is speed; spine refuted as its source; 41% hind foot slip found	1	+71	-0
49c2e55	2026-09-05	Session 3e: refute gearing and contact softness; correct my own stride-length framing	1	+79	-0
7e752e7	2026-09-06	Session 3f: CMA-ES solves the limb-phase gate; 6/6 appears unreachable	19	+9,316,691	-0
9655d79	2026-09-06	Session 3g: withdraw the CMA-ES limb-phase claim; it optimised the wrong metric	7	+3,104,060	-0
8503e04	2026-09-06	Session 3h: fit harness now scores through realism_metrics; 12s fit rejected as unstable	9	+12,601	-3

HASH	DATE	SUBJECT	FILES	+	-
ccebd20	2026-09-06	Session 3i: gate-count objective and the 20 s fit evidence; searches stopped	5	+6,504	-4
2c474f7	2026-09-06	Add docs/HANDOFF.md: complete record of the research and build phases	1	+106	-0
8c7000d	2026-09-06	Session 4: train the walker; the forefoot gate is unreachable on this body	25	+8,558	-38
e184c1e	2026-09-07	Session 4b: drive the sprawl DOFs, couple the tail to the hindlimb	14	+1,552,705	-48
33be60c	2026-09-07	Session 4c: the no-cheat world, and prey the walker cannot catch	11	+1,490	-9
b573e57	2026-09-07	Session 4d: finish the no-cheat world in the environment that has eyes	6	+291	-25
306dd23	2026-09-07	Session 5: hypothalamus -- hunger from the animal's own metabolism	6	+1,100	-2
8b5f491	2026-09-07	Session 5b: fatigue from the published endurance curve; fear is on the wrong sense	6	+294	-3
ef75601	2026-09-07	Session 6: basal ganglia built from equations; NOT ACCEPTED, sweep fails	5	+891	-0
af1b08e	2026-09-07	Session 6b: found the GPR parameters; two of my three errors were structural	5	+340	-141
4a91782	2026-09-07	Session 6c: the living failure map, and 6b's parameters were a variant's	7	+942	-94
cbc42d8	2026-09-07	Session 6d: the basal-ganglia failure was not a failure	4	+327	-30
04b43be	2026-09-07	Session 6e: the winner did not saturate because we were testing the wrong model	8	+2,746	-12
427a845	2026-09-08	Session 6f: the gecko moved onto the validated selector, and brains 1 and 2 run together	4	+505	-6
ba460cb	2026-09-08	Docs: the map gains a flow and a success column	2	+113	-0
6c99f92	2026-09-08	Session 7: the cord becomes an oscillator, so a command has somewhere to land	2	+442	-0
78bcc15	2026-09-08	Session 7 docs: ledger 99-102, the new state, and the paused marker	1	+47	-9
683e8f6	2026-09-08	Session 7b-7c: the dopamine mechanism was wrong the OTHER way, and brain 3 lands	13	+1,553	-575
b9fafcf	2026-09-08	Session 8: scouting the eye found a detector that could not see its own food	4	+425	-15
5158d45	2026-09-08	Session 8b: the retina keeps position, and the published optokinetic asymmetry	6	+978	-12
e4196d8	2026-09-08	Session 8c: the tectum, shipped NOT ACCEPTED because it cannot find real prey	6	+576	-16
034a78d	2026-09-08	Session 9: the literature refuted the plan, and three checks were guarding nothing	30	+18,184	-53

HASH	DATE	SUBJECT	FILES	+	-
ead740b	2026-09-08	Session 9b: the oracle measured, the body hash resolved, and a strike that works	12	+1,140	-13
871f843	2026-09-08	Session 9c: a clip you can actually watch, and two filming artefacts removed	4	+227	-2
3148180	2026-09-08	Session 9d: the gecko was never broken, and the shake was my camera	8	+240	-3
fb64aa2	2026-09-08	Session 9e: the replay camera pointed at the animal's face	12	+541	-5
d017b18	2026-09-08	Session 9f: the lag was a frame-rate header, not the animal	5	+35	-10
010f82c	2026-09-09	Session 9g: rendering the accepted walker side-on, and what the earlier clip was	1	+27	-0
9a7db0a	2026-09-09	Session 9h: three findings from trying to reproduce the Session 4 walking clip	5	+228	-6
3fb0302	2026-09-09	Session 9i: the gait profile is a parameter now, and I retract #179	3	+98	-2
db0163f	2026-09-09	Session 9j: the Session 4 commit and today render the same walk, pixel for pixel	1	+25	-0
779b03c	2026-09-09	Session 9k: the shake was the camera watching the trunk bob	1	+3	-2
7ae5b42	2026-09-09	Session 9l: measuring the push-then-drag rhythm, and rendering the old commit	1	+15	-0
73a37c8	2026-09-09	Session 9m: the user was right -- I filmed the wrong gait profile all session	2	+24	-3
ed61f25	2026-09-09	Session 9n: every clip I sent was the walker this project rejected	3	+186	-3